

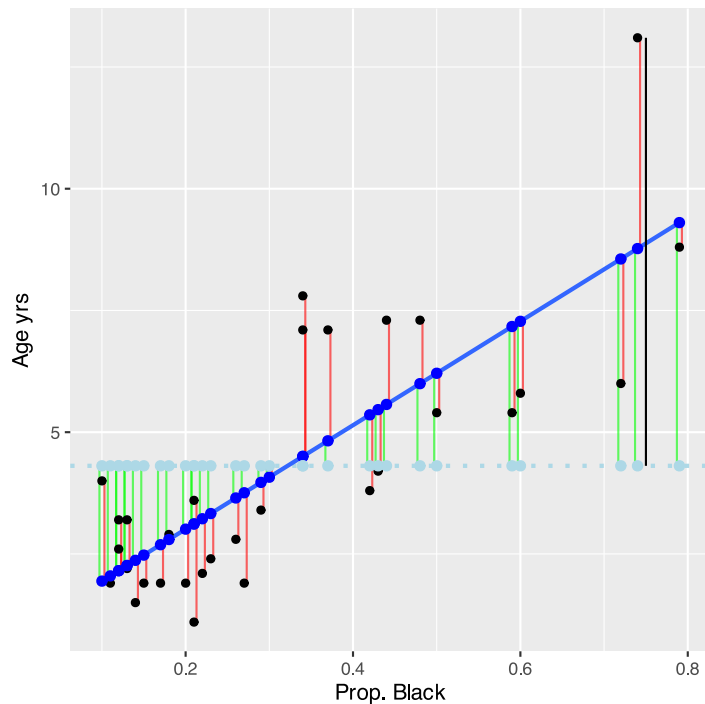
Lecture 10 - Multiple Regression

Bill Perry

Lecture 09: Review

Covered

- Regression T-Test Anova
- Regression Assumptions
- sum of squared allocation and subdivision from total sums of squares



Lecture 10: Overview

Multiple Linear Regression model

- Regression parameters
- Analysis of variance
- Null hypotheses
- Explained variance
- Assumptions and diagnostics
- Collinearity
- Interactions
- Dummy variables
- Model selection
- Importance of predictors

Lecture 10: Analyses

What if more than one predictor (X) variable?

- If predictors continuous
- Mix between categorical and continuous
- Can use multiple linear regression

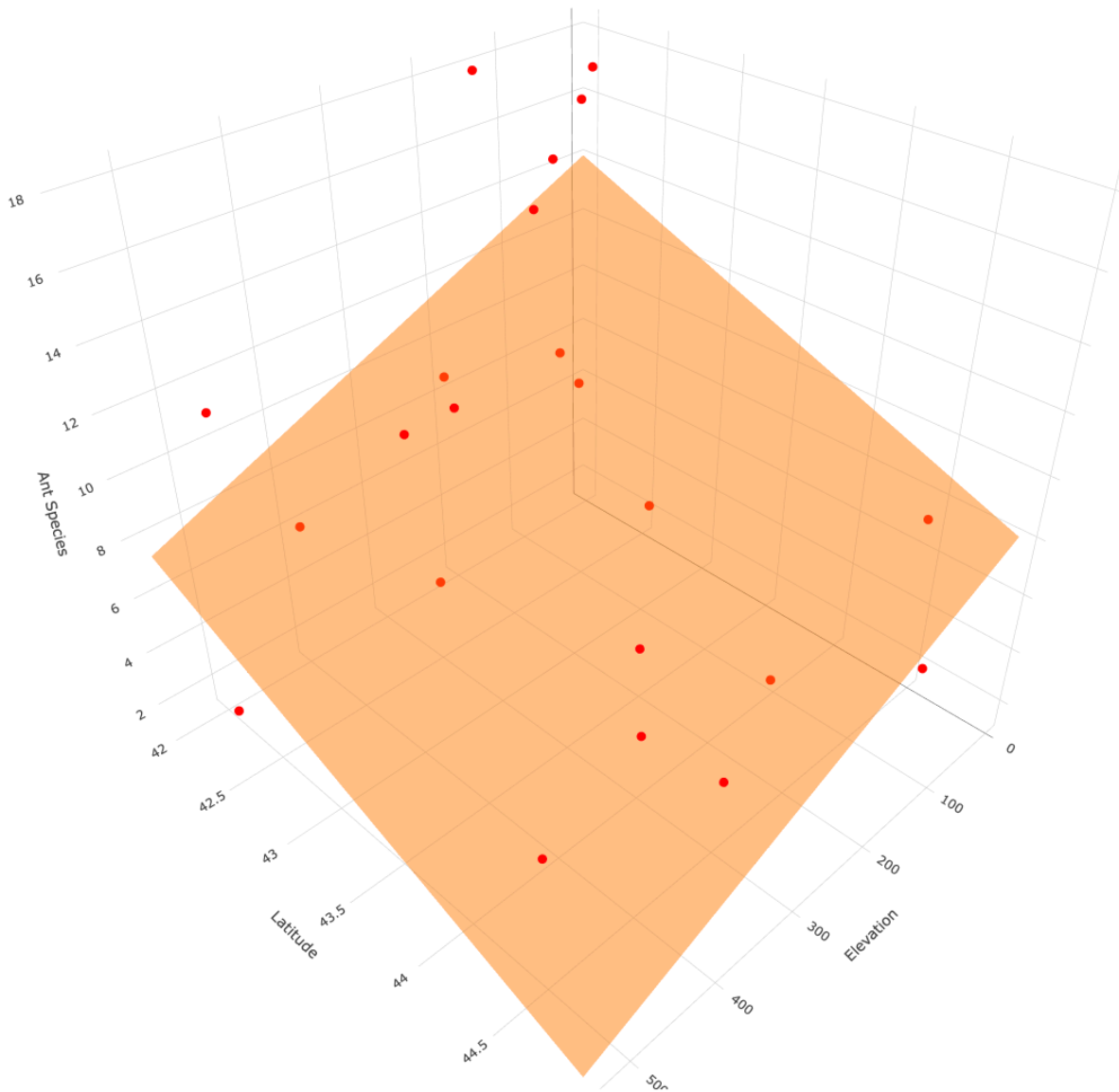
	Independent variable	
Dependent variable	Continuous	Categorical
Continuous	Regression	ANOVA
Categorical	Logistic regression	

Lecture 10: Analyses

Abundance of ants can be modeled as function of

- latitude
- longitude
- both

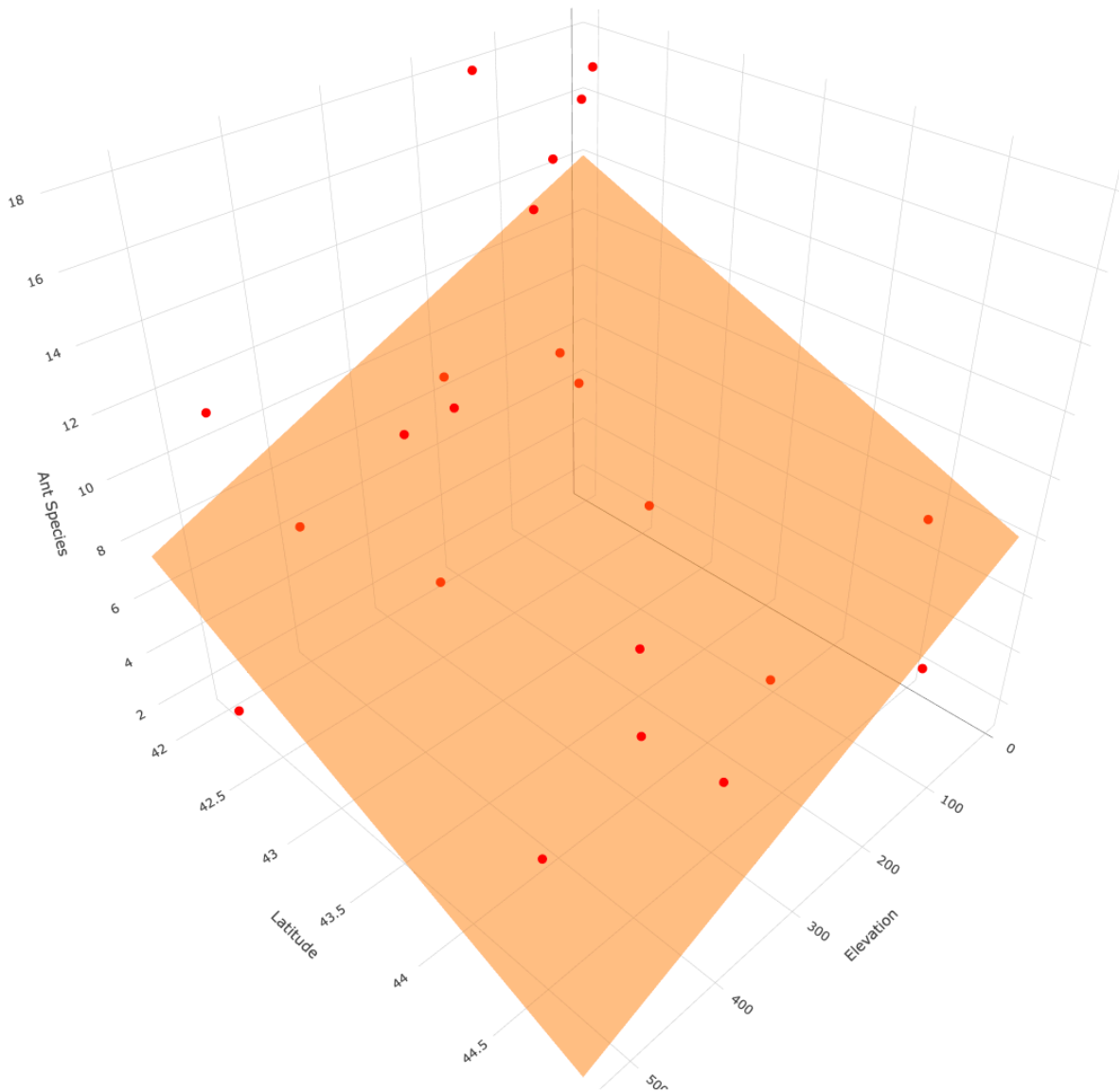
Instead of line, modeled with (hyper)plane



Lecture 10: Analyses

Used in similar way to simple linear regression:

- Describe nature of relationship between Y and X's
- Determine explained / unexplained variation in Y
- Predict new Ys from X
- Find the “best” model



Lecture 10: Analyses

Crawley 2012: “Multiple regression models provide some of the most profound challenges faced by the analyst”:

- Overfitting
- Parameter proliferation
- Multicollinearity
- Model selection



Lecture 10: Analyses

Multiple Regression:

- Set of $i = 1$ to n observations
- fixed X -values for p predictor variables ($X_1, X_2 \dots X_p$)
- random Y -values:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- y_i : value of Y for i^{th} observation $X_1 = x_{i1}, X_2 = x_{i2}, \dots, X_p = x_{ip}$
- β_0 : population intercept, the mean value of Y when $X_1 = 0, X_2 = 0, \dots, X_p = 0$

Lecture 10: Multiple linear regression model

Multiple Regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- β_1 : partial regression slope, change in Y per unit change in X_1 holding other X -vars constant
- β_2 : partial regression slope, change in Y per unit change in X_2 holding other X -vars constant
- β_p : partial regression slope, change in Y per unit change in X_p holding other X -vars constant

What is partial - the relationship between a predictor variable and the response variable **while holding all other predictor variables constant**. It tells you the isolated effect of one variable, controlling for the influence of others.

Lecture 10: Partial regression slopes

model: `ant_spp ~ elevation + latitude`

- The partial slope for elevation tells you how ant species richness changes with elevation when latitude is held constant
 - 1-m increase in elevation - ant spp decrease by 0.012 spp, holding latitude constant
 - Or 100-meter increase in elevation, lose 1.2 spp
- The partial slope for latitude tells you how ant species richness changes with latitude when elevation is held constant
 - 1-degree increase in latitude, ant spp decrease by 2 species, holding elevation constant
 - Moving north = fewer ant spp, even when comparing sites at same elevation

```
model <- lm(ant_spp ~ elevation + latitude, data = ant_df)
summary(model)
```

```
Call:
lm(formula = ant_spp ~ elevation + latitude, data = ant_df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.1180 -2.3759  0.3218  1.9070  5.8369

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  98.49651    26.50701     3.716  0.00147 **
elevation    -0.01226     0.00411    -2.983  0.00765 **
latitude     -2.00981     0.61956    -3.244  0.00427 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.022 on 19 degrees of freedom
Multiple R-squared:  0.5543,    Adjusted R-squared:  0.5074
F-statistic: 11.82 on 2 and 19 DF,  p-value: 0.000463
```

Lecture 10: Regression parameters

Multiple Regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- ϵ_i : unexplained error - difference between y_i and value predicted by model ($\hat{y}_i$)
- $\text{ant_spp} = \beta_0 + \beta_1(\text{lat}) + \beta_2(\text{elevation}) + \epsilon_i$

Lecture 10: Regression parameters

Multiple Regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i$$

- Estimate multiple regression parameters (intercept, partial slopes) using OLS to fit the regression line
- OLS minimize $\sum (y_i - \hat{y}_i)^2$, the SS (vertical distance) between observed y_i and predicted $\hat{y}_i$ for each x_{ij}
- ϵ estimated as residuals: $\epsilon_i = y_i - \hat{y}_i$
- Calculation solves set of simultaneous normal equations with matrix algebra

Lecture 10: Regression parameters

Regression equation can be used for prediction by subbing new values for predictor (X) variables

- Confidence intervals calculated for parameters
- Confidence and prediction intervals depend on number of observations and number of predictors
 - More observations decrease interval width
 - More predictors increase interval width
- Prediction should be restricted to within range of X variables

Lecture 10: Analyses of variance or sums of squares

Variance - SS_{total} partitioned into $SS_{\text{regression}}$ and SS_{residual}

- $SS_{\text{regression}}$ is variance in Y explained by model
- SS_{residual} is variance not explained by model

Source of variation	SS	df	MS	Interpretation
Regression	$\sum_{i=1}^n (y_i - \bar{y})^2$	p	$\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{p}$	Difference between predicted observation and mean
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - p - 1$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1}$	Difference between each observation and predicted
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$		Difference between each observation and mean

Lecture 10: Analyses

SS converted to non-additive MS=(SS/df)

- MS_{residual} : estimate population variance
- $MS_{\text{regression}}$: estimate population variance + variation due to strength of X-Y relationships
- MS is an estimate of variance that doesn't increase with sample size the way Sum of Squares (SS) does

Source of variation	SS	df	MS
Regression	$\sum_{i=1}^n (y_i - \bar{y})^2$	p	$\frac{\sum_{i=1}^n (y_i - \bar{y})^2}{p}$
Residual	$\sum_{i=1}^n (y_i - \hat{y}_i)^2$	$n - p - 1$	$\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - p - 1}$
Total	$\sum_{i=1}^n (y_i - \bar{y})^2$	$n - 1$	

Lecture 10: Hypotheses

Two Null Hypotheses usually tested in MLR:

- "Basic" H_0 : all partial regression slopes equal 0; $\beta_1 = \beta_2 = \dots = \beta_p = 0$
- If "basic" H_0 true, $MS_{\text{regression}}$ and MS_{residual} estimate variance and their ratio (F-ratio) = 1
- If "basic" H_0 false (at least one $\beta \neq 0$) $MS_{\text{regression}}$ estimates variance + partial regression slope and their ratio (F-ratio) will be > 1 - F-ratio compared to F-distribution for p-value

Lecture 10: Hypotheses

Also: is any specific $\beta = 0$ (explanatory role)?

- E.g., does LAT have effect on Ant species?
- These H 's tested through model comparison
- Model 1 - model with 2 predictors X_1, X_2 :
 - $y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \varepsilon_i$
- Model 2 - To test H_0 that $\beta_2 = 0$ compare fit of **model 1** to **model 2** with 1 predictor:
 - $y_i = \beta_0 + \beta_1 X_{i1} + \varepsilon_i$

Lecture 10: Hypotheses

- If $SS_{\text{regression}}$ of model₁=model₂, cannot reject H_0 $\beta_1 = 0$
 - adding X_2 did not explain any additional variance

- If $SS_{\text{regression}}$ of mod1 > mod2, evidence to reject $H_0: \beta_1 = 0$
 - adding X_2 explains more variance
 - evidence to reject $H_0: \beta_2 = 0$
- SS for β_1 is $SS_{\text{extra}\beta_1} = \text{Full } SS_{\text{regression}} - \text{Reduced } SS_{\text{regression}}$
 - This quantifies how much additional variance X_2 explains
- Use partial F-test to test $H_0: \beta_1 = 0$:

$$F_{w, n-p} = \frac{MS_{\text{Extra}}}{FULL MS_{\text{Residual}}}$$

Can also use t-test (R provides this value)

Lecture 10: Explained variance

Explained variance (r^2) is calculated the same way as for simple regression:

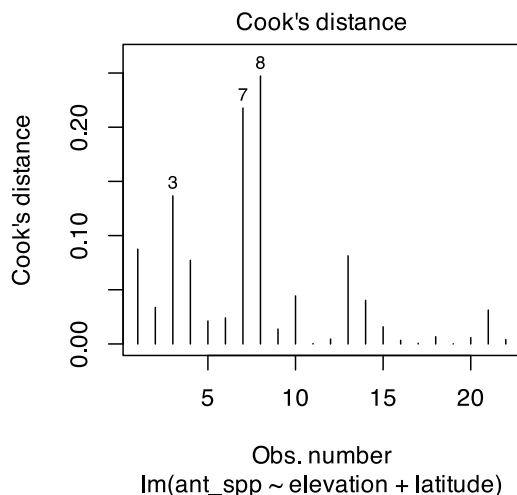
$$r^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}} = 1 - \frac{SS_{\text{Residual}}}{SS_{\text{Total}}}$$

- r^2 values can not be used to directly compare models
- r^2 values will always increase as predictors added
- r^2 values with different transformation will differ

Lecture 10: Assumptions and diagnostics

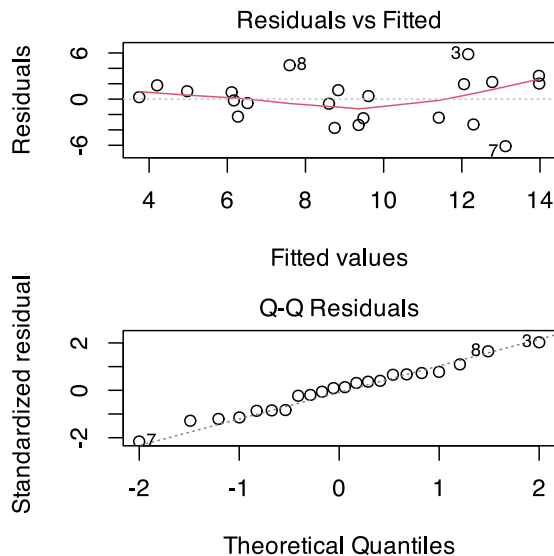
- Assume fixed X s; unrealistic in most biological settings
- No major (influential) outliers
- Check leverage, influence- Cook's Di

```
# Plot Cook's distance
plot(model, which = 4)
```



Lecture 10: Assumptions and diagnostics

- Normality, equal variance, independence
- Residual QQ-plots, residuals vs. predicted values plot
- Distribution/variance often corrected by transforming Y



Lecture 10: Assumptions and diagnostics

- **More observations than predictor variables**
 - Ideally at least 10x observations than predictors to avoid “overfitting” 10:1
 - 2 predictors = 20 observations!!!
- **No colinearity**
 - Need Uncorrelated predictor variables (assessed using scatterplot matrix; VIFs)
- **Each X has linear relationship with Y after accounting for other predictors**
 - **Added Variable (AV) plots** (also called partial regression plots)
 - These show the relationship between Y and X_1 after removing the linear effects of all other predictors
 - Create AV plots for all predictors - `avPlots(model)`
 - Or for just one predictor - `avPlot(model, variable = "proportion_black")`

Lecture 10: Analyses

Regression of Y vs. each X does not consider effect of other predictors:

want to know shape of relationship while holding other predictors constant

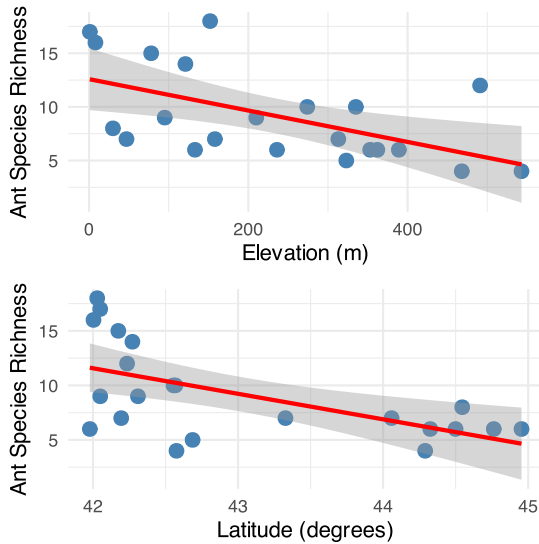
```
#| echo: false
#| message: false
#| warning: false
#| paged-print: false
# Plot 1: Ant species vs Elevation
p1 <- ggplot(ant_df, aes(x = elevation, y = ant_spp)) +
  geom_point(size = 3, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(x = "Elevation (m)",
       y = "Ant Species Richness") +
  theme_minimal()

# Plot 2: Ant species vs Latitude
p2 <- ggplot(ant_df, aes(x = latitude, y = ant_spp)) +
  geom_point(size = 3, color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "red") +
  labs(x = "Latitude (degrees)",
```

```
y = "Ant Species Richness") +  
theme_minimal()
```

```
# Display plots top bottom
```

```
p1 / p2
```



Lecture 10: Collinearity

- Potential predictor variables are often correlated (e.g., morphometrics, nutrients, climatic parameters)
- Multicollinearity (strong correlation between predictors) causes problems for parameter estimates
- Severe collinearity causes unstable parameter estimates: small change in a single value can result in large changes in β - estimates
- Inflates partial slope error estimates, loss of power

	elevation	latitude	ant_spp
elevation	1.0000000	0.1787454	-0.5545244
latitude	0.1787454	1.0000000	-0.5879407
ant_spp	-0.5545244	-0.5879407	1.0000000

Lecture 10: Collinearity

Collinearity can be detected by:

- Variance inflation Factors (VIF):
 - $VIF \text{ for } X_j = 1 / (1 - r^2)$
 - $VIF > 10 = \text{bad}$
- Best/simplest solution:
 - exclude variables that are highly correlated with other variables
 - they are probably measuring similar things and are redundant

Lecture 10: Interactions

Predictors can be modeled as:

- additive (effect of temp, plus precip, plus fertility) or
- multiplicative (interactive)

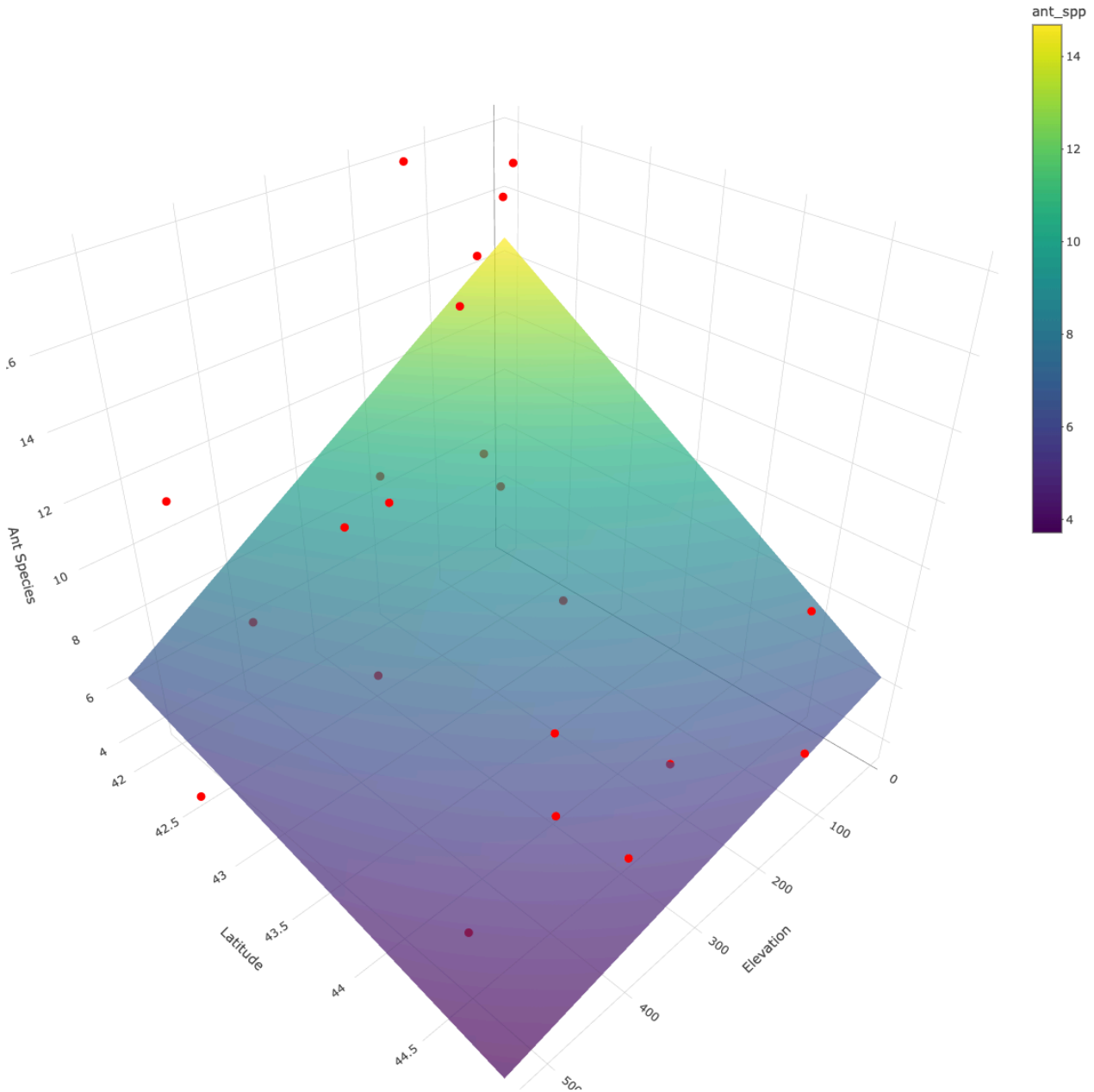
- **Interaction: effect of X_i depends on levels of X_j**
- The partial slope of Y vs. X_1 is different for different levels of X_2 (and vice versa); measured by β_3

$$y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \epsilon_i \quad \text{vs.} \quad y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \beta_3 X_{i1} * X_{i2} \epsilon_i$$

This leads to “Curvature” of the regression (hyper)plane

Lecture 10: Analyses

Ant Species: Interaction Model (Curved Surface)



Lecture 10: Analyses

Adding interactions:

- many more predictors (“parameter proliferation”):
- $2n$ is required; 6 parameters= 64 terms; 7 parameters= 128
- interpretation more complex
- When to include interactions? When they make biological sense!!!

Lecture 10: Dummy variables

Multiple Linear Regression accommodates continuous and categorical variables (gender, vegetation type, etc.) Categorical vars as “dummy vars”, n of dummy variables = n-1 categories

- **Sex M/F:**
 - Need 1 dummy var with two values (0, 1)
- **Fertility L/M/H:**
 - Need 2 dummy var, each with two values (0, 1): fert1 (0 if L or H, 1 if M), fert2 (1 if H, 0 if L or M)

Fertility	fert1	fert2
Low	0	0
Med	1	0
High	0	1

Lecture 10: Analyses

Coefficients interpreted relative to reference condition

- R codes dummy variables automatically
- picks “reference” level alphabetically
- Dummy variables with more than 2 levels add extra predictor variables to model

Fertility	fert1	fert2
Low	0	0
Med	1	0
High	0	1

Lecture 10: Analyses

```
> mod1 <- lm(NPPTotal_gm2y ~ MAT_C + MAP_mm + as.factor(Fertility))
> summary(mod1)
```

Call:
lm(formula = NPPTotal_gm2y ~ MAT_C + MAP_mm + as.factor(Fertility))

Residuals:

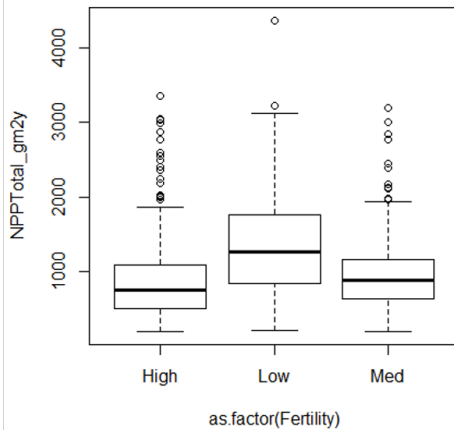
Min	1Q	Median	3Q	Max
-1470.19	-301.59	-49.63	240.82	2901.70

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	359.81714	31.92698	11.270	< 2e-16	***
MAT_C	33.72747	2.73715	12.322	< 2e-16	***
MAP_mm	0.34128	0.03443	9.913	< 2e-16	***
as.factor(Fertility)Low	205.15606	34.36461	5.970	3.1e-09	***
as.factor(Fertility)Med	116.33231	32.30397	3.601	0.000329	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 456.4 on 1232 degrees of freedom
Multiple R-squared: 0.3899, Adjusted R-squared: 0.3879
F-statistic: 196.8 on 4 and 1232 DF, p-value: < 2.2e-16



Lecture 10: Comparing models

- When have multiple predictors (and interactions!)
 - how to choose “best” model?
 - Which predictors to include?
 - Occam’s razor: “best” model balances complexity with fit to data
- To chose:
 - compare “nested” models
- Overfitting
 - getting high r^2 just by having more (useless predictors)
 - so r^2 is not a good way of choosing between nested models

Lecture 10: Comparing models

Need to account for increase in fit with added predictors:

- Adjusted r^2
- Akaike’s information criterion (AIC)
- Both “penalize” models for extra predictors
- Higher adjusted r^2 and lower AIC are better when comparing models
 - p = predictors
 - n = #

$$\text{Adjusted } r^2 = 1 - \frac{SS_{\text{Residual}}/(n - (p + 1))}{SS_{\text{Total}}/(n - 1)}$$

$$\text{Akaike Information Criterion (AIC)} = n[\ln(SS_{\text{Residual}})] + 2(p + 1) - n \ln(n)$$

Lecture 10: Comparing models

- But how to compare models?
 - Can fit all possible models
 - compare AICs or adj- r^2 ,
 - tedious w lots of predictors
 - Automated forward (and backward) stepwise procedures: start w no terms (all terms), add (remove) terms w largest (smallest)
 - partial F statistic

- We will use manual form of backward selection

Lecture 10: Analyses

===== FULL MODEL (Both Variables) =====

```
Call:
lm(formula = ant_spp ~ elevation + latitude, data = ant_df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.1180 -2.3759  0.3218  1.9070  5.8369

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  98.49651    26.50701   3.716  0.00147 **
elevation    -0.01226     0.00411  -2.983  0.00765 **
latitude     -2.00981     0.61956  -3.244  0.00427 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.022 on 19 degrees of freedom
Multiple R-squared:  0.5543,    Adjusted R-squared:  0.5074
F-statistic: 11.82 on 2 and 19 DF,  p-value: 0.000463
```

Lecture 10: Analyses

===== ELEVATION ONLY MODEL =====

```
Call:
lm(formula = ant_spp ~ elevation, data = ant_df)

Residuals:
    Min       1Q   Median       3Q      Max
-4.9010 -2.6947 -0.9502  2.9657  7.6363

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 12.589088    1.385615   9.086 1.55e-08 ***
elevation    -0.014641     0.004913  -2.980  0.0074 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.671 on 20 degrees of freedom
Multiple R-squared:  0.3075,    Adjusted R-squared:  0.2729
F-statistic: 8.881 on 1 and 20 DF,  p-value: 0.0074
```

Lecture 10: Analyses

===== LATITUDE ONLY MODEL =====

```
Call:
lm(formula = ant_spp ~ latitude, data = ant_df)

Residuals:
    Min       1Q   Median       3Q      Max
-6.2223 -2.1188  0.0599  2.1267  6.4990

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 109.8532    30.9803   3.546  0.00203 **
latitude     -2.3401     0.7199  -3.251  0.00401 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.569 on 20 degrees of freedom
Multiple R-squared:  0.3457,    Adjusted R-squared:  0.313
F-statistic: 10.57 on 1 and 20 DF,  p-value: 0.004006
```

Lecture 10: Analyses

===== MODEL COMPARISON TABLE =====

	Model	Adjusted_R2	AIC	R2_vs_Full	AIC_vs_Full
1	Both Variables	0.5074180	115.8646	0.0000000	0.0000000
2	Elevation Only	0.2728722	123.5608	-0.2345459	7.696164
3	Latitude Only	0.3129580	122.3132	-0.1944600	6.448613

===== KEY FINDINGS =====

Full Model AIC: 115.86 (Adjusted R²: 0.5074)

Elevation Only AIC: 123.56 (Adjusted R²: 0.2729)

Latitude Only AIC: 122.31 (Adjusted R²: 0.3130)

--- Differences from Full Model ---

Removing latitude: AIC increases by 7.70, Adj R² decreases by -0.2345

Removing elevation: AIC increases by 6.45, Adj R² decreases by -0.1945

===== CONCLUSION =====

✓ Full model has LOWEST AIC (best fit)

✓ Full model has HIGHEST Adjusted R^2 (explains most variance)

✗ Removing either variable worsens model performance

Both elevation and latitude are important predictors of ant species diversity!

Lecture 10: Predictors

Usually want to know relative importance of predictors to explaining Y

- Three general approaches:
- Using F-tests (or t-tests) on partial regression slopes
- Using coefficient of partial determination
- Using standardized partial regression slopes

Lecture 10: Predictors

Using F-tests (or t-tests) on partial regression slopes:

- Conduct F tests of H_0 that each partial regression slope = 0
- If cannot reject H_0 , discard predictor
- Can get additional clues from relative size of F-values
- Does not tell us absolute importance of predictor (usually can not directly compare slope parameters)

Lecture 10: Predictors

Using coefficient of partial determination:

- the reduction in variation of Y due to addition of predictor (X_j)

$$r_{X_j}^2 = \frac{SS_{\text{Extra}}}{\text{Reduced } SS_{\text{Residual}}}$$

SS_{Extra}

- Increased in $SS_{\text{regression}}$ when X_j is added to model
- Reduced SS_{residual} is the unexplained SS from model without X_j

Lecture 10: Predictors

Using standardized partial regression slopes:

- predictors of predictor variables can not be directly compared
- Why?
 - Standardize all vars (mean = 0, sd= 1)
 - Scales are identical and larger PRS mean more important variable

Lecture 10: Predictors

- Using partial r^2 values
 - **pEta_sqr (Partial Eta Squared):**
 - **Elevation: 0.3189 (31.9%)** - Elevation explains 31.9% of the variance *after controlling for latitude*
 - **Latitude: 0.3564 (35.6%)** - Latitude explains 35.6% of the variance *after controlling for elevation*
 - Both are important predictors! Latitude is slightly stronger
- dR_sqr (Delta R Squared / Semi-partial R^2):
 - Elevation: 0.2087 (20.9%)
 - If you removed elevation from the model, R^2 would drop by 20.9%
 - Latitude: 0.2468 (24.7%)
 - If you removed latitude from the model, R^2 would drop by 24.7%
- Shows the unique contribution of each predictor to the total variance explained

== Method 2: Type II ANOVA (car package) ==

Anova Table (Type II tests)

```
Response: ant_spp
      Sum Sq Df F value    Pr(>F)
elevation  81.224  1  8.8955 0.007652 **
latitude   96.085  1 10.5231 0.004272 **
Residuals 173.487 19
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

== Coefficients Table ==

Coefficients	SSR	df	pEta_sqr	dR_sqr
elevation	81.22	1	0.3189	0.2087
latitude	96.09	1	0.3564	0.2468
Residuals	173.49	19	0.5000	0.4457

== Summary Statistics ==

Sum of squared errors (SSE): 173.5

Sum of squared total (SST): 389.3

Model R^2 : 0.5543